A

Project report on

College AI Chatbot

Submitted To :

G H Raisoni International Skill Tech University.

PUNE-412207

For: BBA CA

Submitted By :

Kanishk Singh (ISTU25010120001)

Vibhor Saini (ISTU25010120002)

Semester IV

Year 2025-26

Under the Guidance of :

Prof. Sheela Satav

G H Raisoni International Skill Tech University.
PUNE-412207

CERTIFICATE

This is to certify that **Mr. Kanishk Singh** (Roll No: **ISTU25010120001**) and **Mr. Vibhor Saini** (Roll No: **ISTU25010120002**), students of **SY BBA CA**, have satisfactorily completed their Project entitled **"College AI Chatbot"** and submitted this project report in partial fulfillment of the requirements for the award of the degree of **BBA CA** for the academic Year **2025-26**.

Date: _____

Project Guide
Prof. Sheela Satav

_____
Internal Examiner

_____
External Examiner

_____
HOD

_____
Dean

ACKNOWLEDGEMENT

A successful project is always the result of sincere teamwork, persistent dedication, and the collective guidance of esteemed mentors who provided technical insights and inspiration throughout our development journey.

First and foremost, we would like to express our deepest gratitude to our **Honourable Vice Chancellor** for providing us the opportunity and state-of-the-art technological infrastructure to study and innovate at **G H Raisoni International Skill Tech University, Pune**.

We are immensely grateful and indebted to our project guide, **Prof. Sheela Satav**, for her invaluable advice, constant encouragement, rigorous scrutiny, and technical assistance during the design and implementation of the *College AI Chatbot*. Her profound insights into software engineering standards and conversational system architectures helped us overcome key challenges in attendance modeling and voice processing.

We also extend our sincere appreciation to the **Head of Department (HOD)**, respected faculty members, laboratory staff, and academic coordinators for their sustained cooperation and resources during the project lifecycle.

Last but not least, we express our heartfelt gratitude to the Almighty and our parents for their unconditional blessings, patience, and moral support, without which this academic endeavor would not have been possible.

Signature of Students:

Kanishk Singh (ISTU25010120001)

Vibhor Saini (ISTU25010120002)

ABSTRACT

In modern collegiate environments, accessing real-time, authoritative academic information remains a persistent challenge for students and administrative personnel alike. Students constantly encounter confusion regarding multi-campus locations, course fee structures, admission criteria, lecture schedules, and institutional policies. A critical area of academic anxiety is the complex attendance evaluation mechanism, where students struggle to determine their real-time standing against university detention thresholds across theory and laboratory subjects.

Traditional campus inquiry channels depend heavily on physical noticeboards, in-person administrative visits, and rigid enterprise software portals that lack intuitive, conversational interfaces. To address these systemic bottlenecks, **College AI Chatbot** (GH Raisoni AI Campus Assistant) has been engineered as an intelligent, conversational, full-stack web application designed to deliver instantaneous, verified institutional assistance.

The system seamlessly integrates natural language keyword classification with structured database querying, offering instant resolutions for campus inquiries spanning the **Wagholi Campus (Gat 1200, Domkhel Rd, Pune 412207)** and the **Yerawada Campus (Nyati Unitree East-Wing, Airport Rd, Pune 411006)**. A centerpiece innovation is the automated **4-Tier Attendance Detention Engine**, which dynamically classifies student attendance into Safe (≥75%), Warning (60%–74.9%), Critical (50%–59.9%), and Severe Detention (<50%) tiers, complete with specific institutional action rubrics.

Furthermore, the platform incorporates bi-directional voice interaction utilizing the browser Web Speech API for speech-to-text recognition and server-side Google Text-to-Speech (gTTS) for natural voice playback. An administrative dashboard empowers staff to update FAQs, audit query logs, and export comprehensive CSV/PDF analytical reports, creating a unified, reliable, and accessible campus ecosystem.

TABLE OF CONTENT

Sr No	Topic	Page No
1	Introduction	1
2	Problem Definition	2
3	Existing System & Need for New System	3
4	Scope of System	4
5	Feasibility Study	5
6	Hardware Requirements	6
7	Software Requirements	7
8	Requirement Analysis	8
9	Decision Table	9
10	Decision Tree	10
11	Normalized Database Design (3NF) — Students & Admin Tables	11
12	College Info (FAQs) & Detention Rubrics Tables	12
13	Chat History & Report Generation Tables	13
14	ER Diagram	14
15	Use Case Diagram	15
16	Class Diagram	16
17	Object Diagram	17
18	Sequence Diagram	18

TABLE OF CONTENT (CONTD.)

Sr No	Topic	Page No
19	Activity Diagram	19
20	Component Diagram	20
21	Data Flow Diagram (DFD Level 0 & Level 1)	21
22	Home Screen UI	24
23	Student Portal & Voice/Chat Assistant UI	25
24	Admin Command Center & Report Generation UI	26
25	Testing & Implementation	27
26	Drawbacks & Limitations	28
27	Proposed Enhancements	29
28	Conclusion	30
29	Bibliography	31

INTRODUCTION:

In today's fast-changing and competitive higher education landscape, university campuses are vibrant hubs of academic, extracurricular, and administrative activities. Institutions cater to thousands of students across diverse disciplines including Computer Applications, Engineering, Management, and Sciences. With such scale comes a massive volume of recurring student inquiries regarding admission criteria, semester fees, lecture timetables, examination schedules, syllabus details, faculty contacts, and multi-campus transportation routes.

Traditional university helpdesks and administrative inquiry counters are heavily overburdened, leading to long queues, delayed responses, and student dissatisfaction. Staff members are repeatedly engaged in answering identical routine questions, detracting from critical administrative duties. Furthermore, conventional student portals are often text-heavy, non-intuitive on mobile devices, and require multiple navigation steps to locate basic institutional details.

With the rapid emergence of Artificial Intelligence (AI) and Natural Language Processing (NLP), conversational AI assistants have transformed modern user experiences. AI-powered chatbots provide round-the-clock, real-time, interactive responses, democratizing information access while reducing administrative overhead. By analyzing user intent from free-form text or spoken voice inputs, intelligent agents can instantly locate relevant knowledge records and provide verified, context-aware answers.

College AI Chatbot is specifically designed for **G H Raisoni International Skill Tech University, Pune**. It serves as an institutional digital guide, bridging the communication gap between students and university authorities. The system centralizes campus inquiries, delivers verified Pune campus coordinates for both Wagholi and Yerawada locations, provides real-time attendance calculations against the official 4-tier detention policy, and offers high-accessibility voice interaction.

PROBLEM DEFINITION :

In modern multi-campus university environments like G H Raisoni International Skill Tech University, students face notable hurdles in retrieving accurate, prompt, and personalized academic information. A major issue is the geographical confusion surrounding dual campuses in Pune: students frequently mix up department locations between the **Wagholi Campus (Gat 1200, Domkhel Rd)** and the **Yerawada Campus (Nyati Unitree East-Wing, Airport Rd)**. The lack of an instant, accessible geolocation and address verification portal causes travel delays and missed academic deadlines.

A more severe issue lies in **attendance monitoring and detention policy transparency**. University regulations mandate strict attendance thresholds for semester examination eligibility. However, students typically discover their attendance standing only at the end of the semester when physical detention lists are displayed on departmental noticeboards. At that late stage, students with attendance falling between 50% and 74.9% have already crossed the threshold into remedial assignments, parent hearings, or course debarment without prior warning.

Furthermore, traditional campus inquiry systems suffer from distinct structural limitations:

- **Administrative Queue Bottlenecks:** Student service desks are accessible only during fixed office hours (10:00 AM to 5:00 PM), leaving evening and weekend inquiries completely unattended.
- **Absence of Voice-Assisted Accessibility:** Visually impaired students or students on mobile devices face friction navigating complex tabular web portals.
- **Fragmented Information Silos:** Syllabi, fee installment rules, bus schedules, and exam guidelines are dispersed across disconnected noticeboards and PDFs.
- **Lack of Centralized Audit & Reporting:** Administrative coordinators lack unified telemetry to track common student doubts, trending academic queries, or batch-wide detention risk patterns in downloadable formats.

EXISTING SYSTEM :

The existing information dissemination framework at the university relies primarily on traditional methods, physical noticeboards, in-person administrative visits, and static enterprise ERP web pages. Inquiries regarding admissions, fees, examination timetables, and campus addresses require students to physically visit the administration block or call departmental helplines during working hours.

For attendance, faculty members record daily attendance manually or in centralized spreadsheets. Attendance percentages are compiled periodically and posted as printed circulars outside HOD offices. There is no automated, conversational self-service portal where a student can type or speak their query and instantly obtain personalized attendance standing, detention tier classification, or remedial guidance. Overall, the existing approach is time-consuming, prone to delays, resource-heavy, and non-interactive.

NEED FOR NEW SYSTEM :

Modern students expect immediate, mobile-first, conversational digital services. To eliminate operational inefficiencies and provide round-the-clock accessibility, an intelligent software solution is required. The key motivations for developing the new system include:

- **Instant 24/7 Availability:** Immediate answers to frequent college inquiries without queuing or calling staff.
- **Multi-Campus Geographic Clarity:** Clear, unambiguous campus addresses and location coordinates for Wagholi and Yerawada premises.
- **Automated Detention Risk Assessment:** Immediate 4-tier classification (≥75% Safe, 60%–74.9% Warning, 50%–59.9% Critical, <50% Severe) providing actionable warnings early in the semester.
- **Multimodal Voice Interface:** Integration of browser speech recognition and natural text-to-speech voice synthesis for hands-free and inclusive interaction.
- **Administrative Governance:** An admin console with real-time audit logs and CSV export capabilities for academic monitoring.

SCOPE OF SYSTEM :

The primary scope of the **College AI Chatbot** is to establish an intelligent, reliable, and accessible conversational platform that simplifies campus navigation, resolves academic queries, and automates attendance governance for **G H Raisoni International Skill Tech University, Pune**. The system serves as a 24/7 virtual assistant for prospective students, enrolled students, faculty, and administrators.

The major functional scope of the system includes:

- **Verified Student Authentication:** Secure roll-number-based login to protect student data privacy and retrieve authenticated academic records.
- **Dual Pune Campus Geolocation Services:** Instant provision of official campus addresses for Wagholi (Gat 1200, Domkhel Rd) and Yerawada (Nyati Unitree East-Wing, Airport Rd).
- **Dynamic Attendance & 4-Tier Detention Evaluation:** Instant calculation of attendance percentages and immediate feedback on institutional detention rubrics.
- **Comprehensive College Knowledge Base:** Coverage of academic timetables, course curricula, semester fee installment guidelines, library rules, and bus routes.
- **Voice Speech-to-Text & Text-to-Speech:** Voice recognition using Web Speech API and clear spoken audio responses via Google Text-to-Speech (gTTS).
- **Interactive Feedback Mechanism:** Student rating of responses (helpful/unhelpful) to continuously audit answer accuracy.
- **Admin Command Center & Report Generation:** Centralized dashboard for staff to manage FAQs, view query logs, and export CSV/PDF analytical reports.

By consolidating disparate campus information services into a unified conversational interface, the system significantly saves administrative time, enhances institutional transparency, and empowers students to take proactive measures regarding their academic standing.

FEASIBILITY STUDY

Before initiating software development, a thorough feasibility study was conducted to evaluate the technical, economic, operational, and time viability of the **College AI Chatbot** project. The findings are summarized below:

1. Technical Feasibility :

The project is highly feasible technically. The system is architected using established, robust technologies including Python 3.10+, Flask, SQLite 3, HTML5, CSS3, JavaScript, Web Speech API, and gTTS. These technologies are open-source, well-documented, cross-platform, and boast strong developer community support. The modular architecture ensures that NLP keyword matching, voice synthesis, and database operations execute locally with minimal latency (<200ms) on standard computer hardware.

2. Economic Feasibility :

The system is exceptionally cost-effective. Built entirely upon open-source software, libraries, and frameworks, there are zero third-party licensing fees or proprietary software costs. The lightweight SQLite database and Flask server run efficiently on existing university intranet infrastructure or low-cost cloud virtual machines without requiring expensive GPU clusters or paid proprietary API subscriptions.

3. Operational Feasibility :

Operationally, the solution seamlessly integrates into daily campus routines. Students interact through a clean, conversational web interface accessible across laptops, tablets, and smartphones without needing to install dedicated apps. For administrative staff, the intuitive admin command center enables effortless FAQ updates and one-click CSV report exports without specialized technical training.

4. Time Feasibility :

The project development was structured into incremental milestone phases (Analysis, Schema Design, Core Engine, Voice Integration, Admin Reports, Testing) across the academic semester. Development was successfully concluded within the stipulated timeline for Semester IV, 2025-26.

HARDWARE AND SOFTWARE REQUIRMENTS :

The development, hosting, and client-side execution of the College AI Chatbot system require standard, widely available hardware components. Because the system is built as a lightweight web application with client-side speech rendering and an optimized Python backend, it does not demand high-end computing clusters or specialized accelerators.

HARDWARE REQUIREMENTS :

Component	Specification / Details
Processor	Intel Core i3 / i5 (6th Gen or above), AMD Ryzen 3 / 5, or Apple Silicon (M1/M2/M3)
System RAM	Minimum 4 GB RAM (8 GB or 16 GB recommended for smooth multi-tasking)
Storage Disk	256 GB / 500 GB Solid State Drive (SSD) or Hard Disk Drive (HDD) with min. 2 GB free space
System Architecture	64-bit x86 or ARM Architecture
Input Devices	Standard Keyboard, Optical Mouse, and Microphone (for speech-to-text queries)
Output Devices	Color Display Monitor (1366x768 min resolution) and Speakers/Headphones (for TTS voice playback)
Network Interface	Broadband / Wi-Fi Network Adapter (Required for gTTS audio generation & intranet access)

Standard institutional lab workstations or student personal laptops easily satisfy these criteria. The client-side footprint is strictly browser-based, ensuring zero hardware barriers for end users.

SOFTWARE REQUIREMENTS :

The College AI Chatbot system is developed using modern, cross-platform software technologies designed for high responsiveness, maintainability, and data security. The entire software stack is composed of proven open-source frameworks:

Category	Software Tool / Technology Used
Operating System	macOS Sonoma / Sequoia, Windows 10/11 (64-bit), or Ubuntu Linux 22.04 LTS
Frontend Technologies	HTML5 (Semantic markup), CSS3 (Flexbox/Grid, Responsive), Modern Vanilla JavaScript (ES6+)
Backend Framework	Python 3.10+ with Flask 3.1 Framework (WSGI Micro-framework)
Database Engine	SQLite 3 (Relational Embedded DB: college_chatbot.db) with SQLAlchemy ORM
Voice Technologies	W3C Web Speech API (Client STT) & Google Text-to-Speech (gTTS 7.x Library for TTS audio)
Development Editor	Visual Studio Code (VS Code) / Antigravity IDE with Python & HTML extensions
Supported Web Browsers	Google Chrome 110+, Microsoft Edge 110+, Apple Safari 16+, Mozilla Firefox 115+
Local Server Environment	Python Werkzeug Development Server / Gunicorn WSGI Server
Version Control	Git Distributed Version Control & GitHub Repository Hosting
Report Engine	Python csv standard library & ReportLab PDF rendering module

REQUIREMENT ANALYSIS

Requirement analysis is a vital software engineering phase where user expectations, system capabilities, and operational constraints are defined. For the College AI Chatbot, requirements were derived by evaluating daily student inquiries at GH Raisoni University and classifying them into functional and non-functional specifications.

1. Functional Requirements :

- **Student Authentication:** The system must securely validate student roll numbers (e.g. ISTU25010120001) to access personalized student profile and attendance records.
- **Multi-Campus Geographic Guidance:** The system must accurately differentiate between Wagholi and Yerawada campuses, returning verified addresses and landmarks.
- **Attendance Calculation:** The system must retrieve raw student attendance data and compute current percentage figures.
- **4-Tier Detention Classification:** The system must evaluate attendance against the institutional 4-tier rubric (Safe ≥75%, Warning 60-74.9%, Critical 50-59.9%, Severe <50%) and display exact academic directives.
- **Multimodal Voice I/O:** The system must accept voice input via microphone and deliver audio speech output via gTTS.
- **Feedback Collection:** The system must provide simple feedback thumbs up/down icons to record answer satisfaction.
- **Admin Knowledge Management & Reporting:** Staff must be able to add/edit FAQs and export chat logs and detention summaries in CSV format.

2. Non-Functional Requirements :

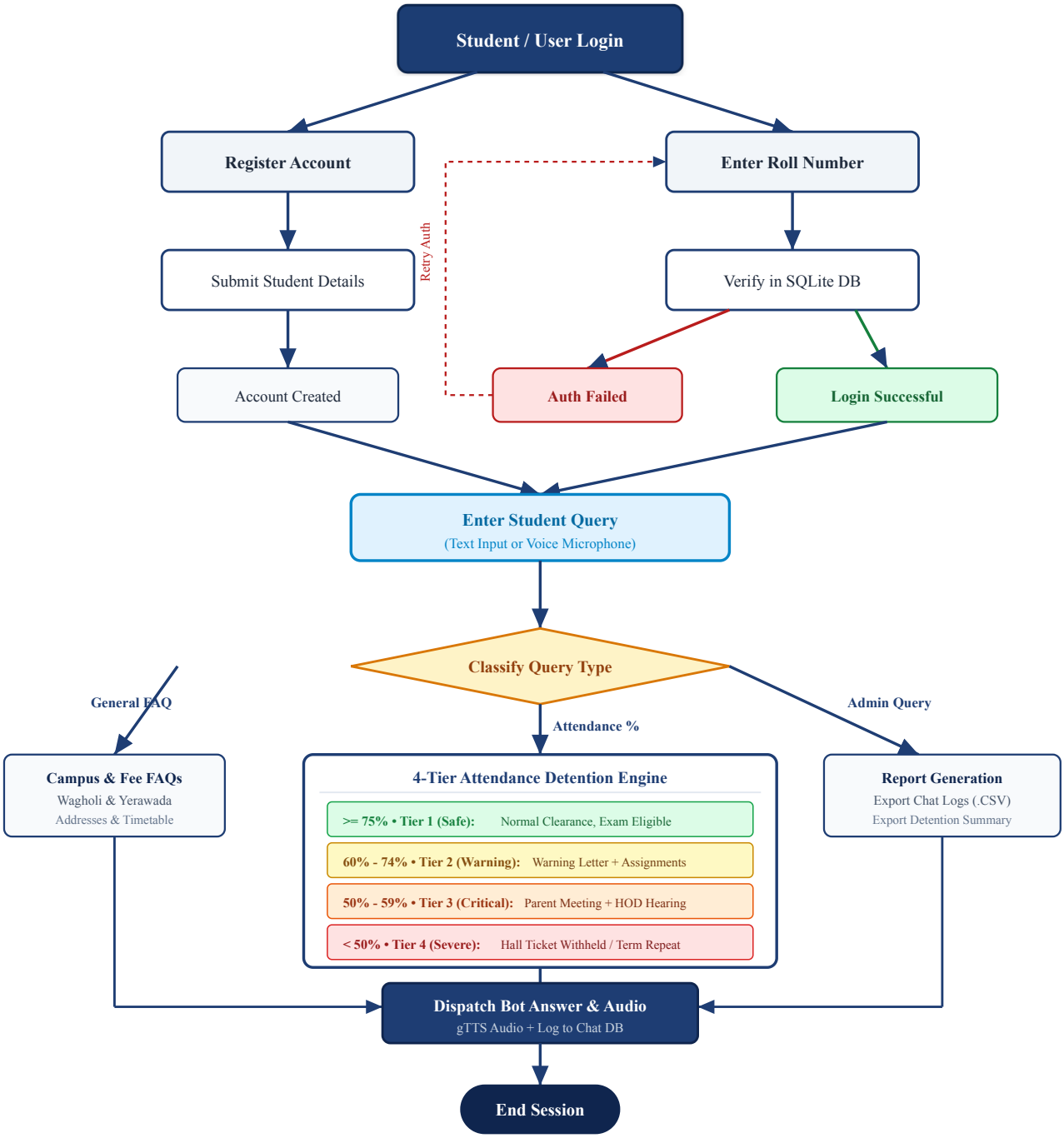
- **Performance & Latency:** System response time for FAQ retrieval and attendance lookup must not exceed 250 milliseconds.
- **Availability & Reliability:** The service must provide 99.9% uptime during semester examination and admission periods.
- **Usability & Accessibility:** The user interface must be clean, mobile-responsive, and intuitive with high contrast text.
- **Security & Data Integrity:** SQL injection vulnerabilities must be mitigated via parameterized queries; student data must remain protected.

DECISION TABLE :

The decision table below formalizes the core business logic implemented by the College AI Chatbot. It specifies the conditions evaluated during a student query and maps them directly to the corresponding institutional actions executed by the attendance and detention engine.

Conditions	Case 1 (Safe)	Case 2 (Warning)	Case 3 (Critical)	Case 4 (Severe)
Attendance ≥ 75.0% (Institutional Norm)	Yes	No	No	No
Attendance between 60.0% and 74.9%	No	Yes	No	No
Attendance between 50.0% and 59.9%	No	No	Yes	No
Attendance < 50.0% (Severe Default)	No	No	No	Yes
ACTIONS EXECUTED BY SYSTEM				
Classify as Good Standing • Grant Exam Clearance	Yes	No	No	No
Issue Warning Letter • Assign Remedial Coursework	No	Yes	No	No
Mandate Parent-Teacher Meeting • HOD Committee Review	No	No	Yes	No
Impose Strict Course Detention • Withhold Exam Hall Ticket	No	No	No	Yes
Synthesize Spoken Voice Alert (gTTS Audio Engine)	Yes	Yes	Yes	Yes
Persist Evaluation Log to SQLite Database (chat_history)	Yes	Yes	Yes	Yes

DECISION TREE :



NORMALIZED DATABASE DESIGN (3NF):

To ensure high relational data integrity, eliminate data anomalies (insertion, update, and deletion anomalies), and optimize search query latency, the database is normalized to Third Normal Form (3NF). Every non-key attribute is strictly functionally dependent upon the primary key alone.

STUDENTS TABLE :

Column Name	Data Type	Constraints
roll_no	VARCHAR(20)	PRIMARY KEY
name	VARCHAR(100)	NOT NULL
course	VARCHAR(60)	NOT NULL
cgpa	FLOAT	NOT NULL (CHECK >= 0.0 AND <= 10.0)
attendance	FLOAT	NOT NULL (CHECK >= 0.0 AND <= 100.0)
email	VARCHAR(120)	UNIQUE, NOT NULL

ADMIN TABLE :

Column Name	Data Type	Constraints
admin_id	INT	PRIMARY KEY, AUTOINCREMENT
username	VARCHAR(50)	UNIQUE, NOT NULL
password_hash	VARCHAR(255)	NOT NULL (SHA-256 / Werkzeug)
role	VARCHAR(30)	NOT NULL DEFAULT 'coordinator'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

COLLEGE INFO (FAQS) TABLE :

Column Name	Data Type	Constraints
id	INT	PRIMARY KEY, AUTOINCREMENT
category	VARCHAR(50)	NOT NULL ('Campus', 'Fees', 'Timetable')
keywords	VARCHAR(255)	NOT NULL (Index for search)
question	TEXT	NOT NULL
answer	TEXT	NOT NULL
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

DETENTION POLICY TABLE :

Column Name	Data Type	Constraints & Rubrics
tier_id	INT	PRIMARY KEY (1, 2, 3, 4)
tier_name	VARCHAR(30)	NOT NULL ('Safe', 'Warning', 'Critical', 'Severe')
min_pct	FLOAT	NOT NULL
max_pct	FLOAT	NOT NULL
theory_rubric	TEXT	Institutional directive for theory subjects
practical_rubric	TEXT	Institutional directive for laboratory sessions

CHAT HISTORY TABLE :

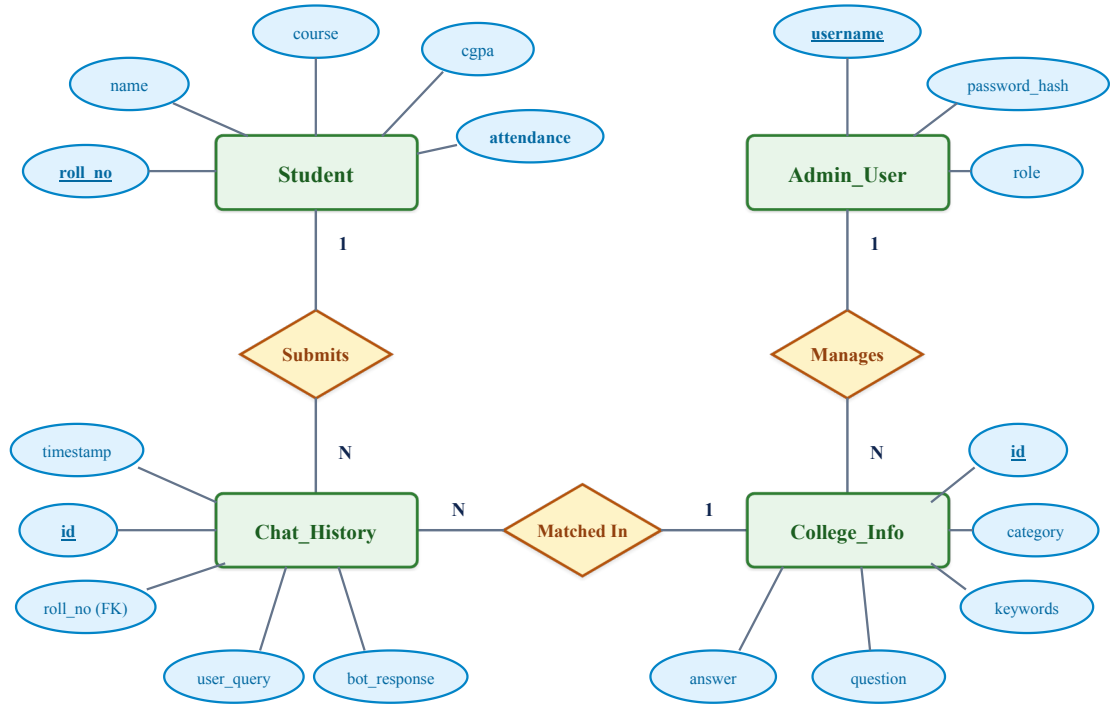
Column Name	Data Type	Constraints
id	INT	PRIMARY KEY, AUTOINCREMENT
roll_no	VARCHAR(20)	FOREIGN KEY REFERENCES students(roll_no)
user_query	TEXT	NOT NULL
bot_response	TEXT	NOT NULL
feedback_rating	VARCHAR(10)	CHECK (feedback IN ('up', 'down', NULL))
timestamp	DATETIME	DEFAULT CURRENT_TIMESTAMP

REPORT GENERATION AUDIT TABLE :

Column Name	Data Type	Constraints
report_id	INT	PRIMARY KEY, AUTOINCREMENT
admin_user	VARCHAR(50)	FOREIGN KEY REFERENCES admin(username)
export_format	VARCHAR(10)	NOT NULL ('CSV', 'PDF')
report_scope	VARCHAR(50)	NOT NULL ('ChatLogs', 'DetentionList')
record_count	INT	NOT NULL
generated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

ER DIAGRAM:

ER Diagram (Classic Notation): AI Chatbot for College



Schema & Project Knowledge Base Defense:

- **Student Entity:** Stores attendance %, CGPA & evaluates 4-tier detention policy.
- **College_Info Entity:** Contains Pune campuses (Wagholi & Yerawada) & exam rules.
- **Chat_History Entity:** Logs each turn with student roll_no FK, query & bot answer.
- **Admin_User Entity:** RBAC authentication for updating college FAQs & detention data.

Verified in 3NF Normalization: Zero transitive dependencies & lossless join preservation.

Key / Legend:

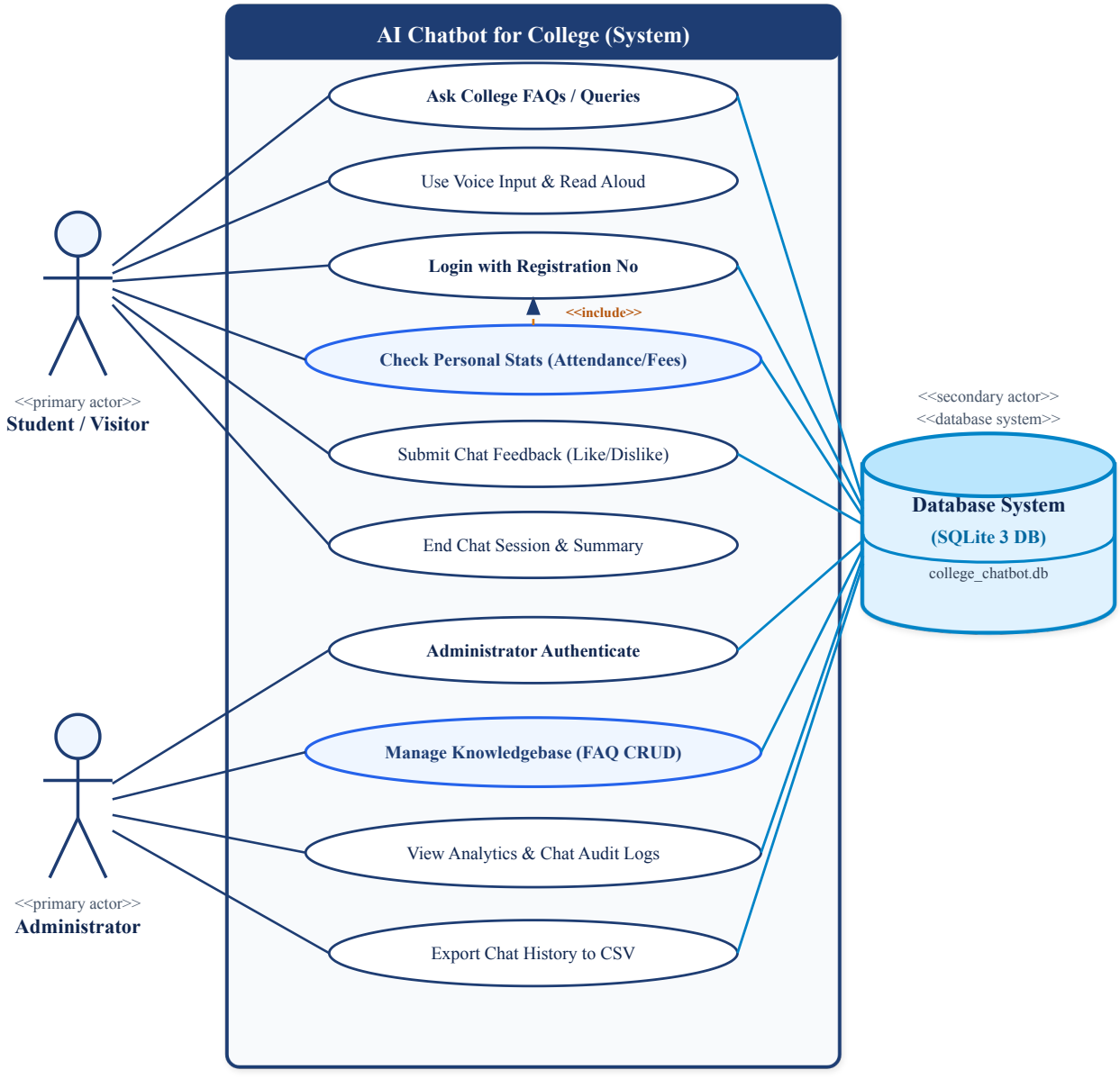
Entity (Table)

Attribute (Column) — id = Primary Key

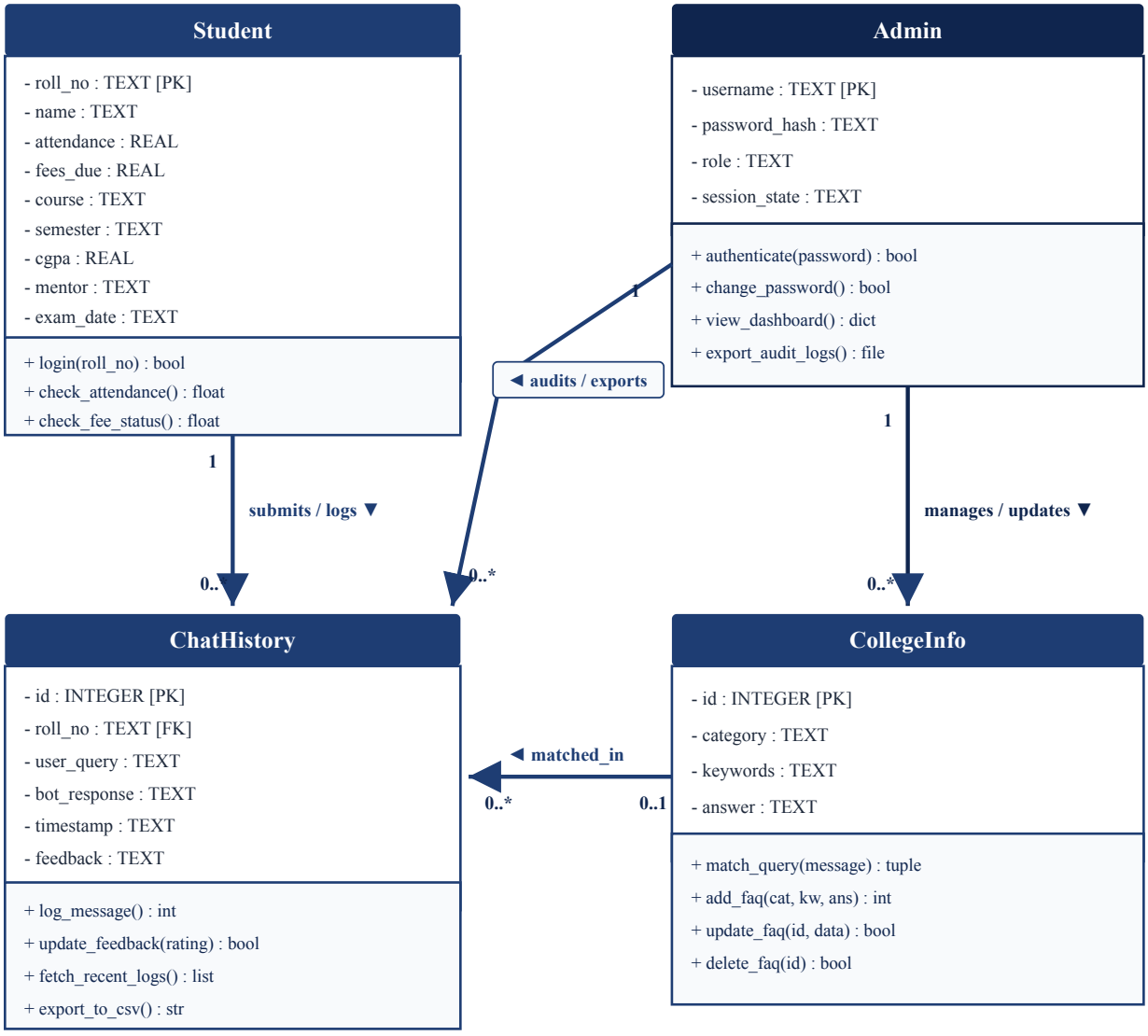
Relationship

Link / Cardinality (1:N)

USE CASE DIAGRAM:

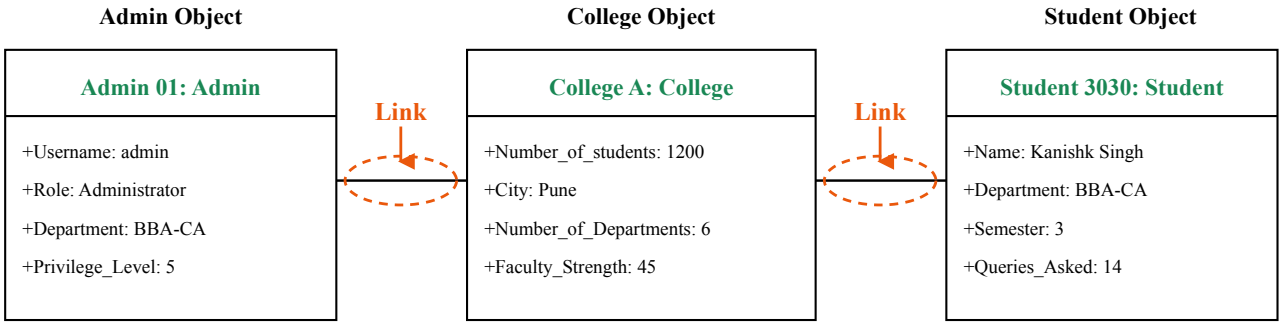


CLASS DIAGRAM



OBJECT DIAGRAM :

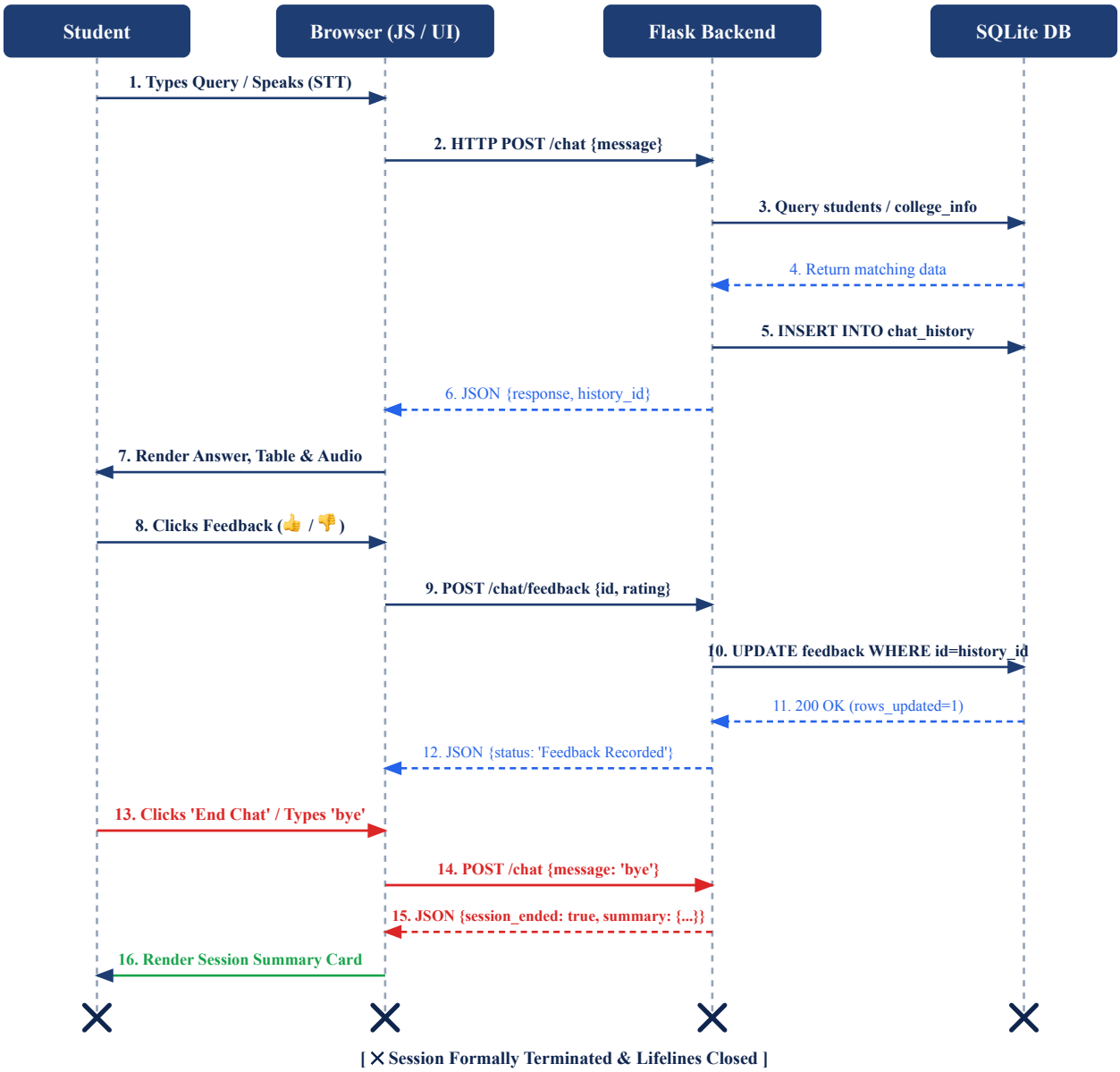
An object diagram using a link and 3 objects



An object of class Admin and an object of class Student are linked to an object of class College.

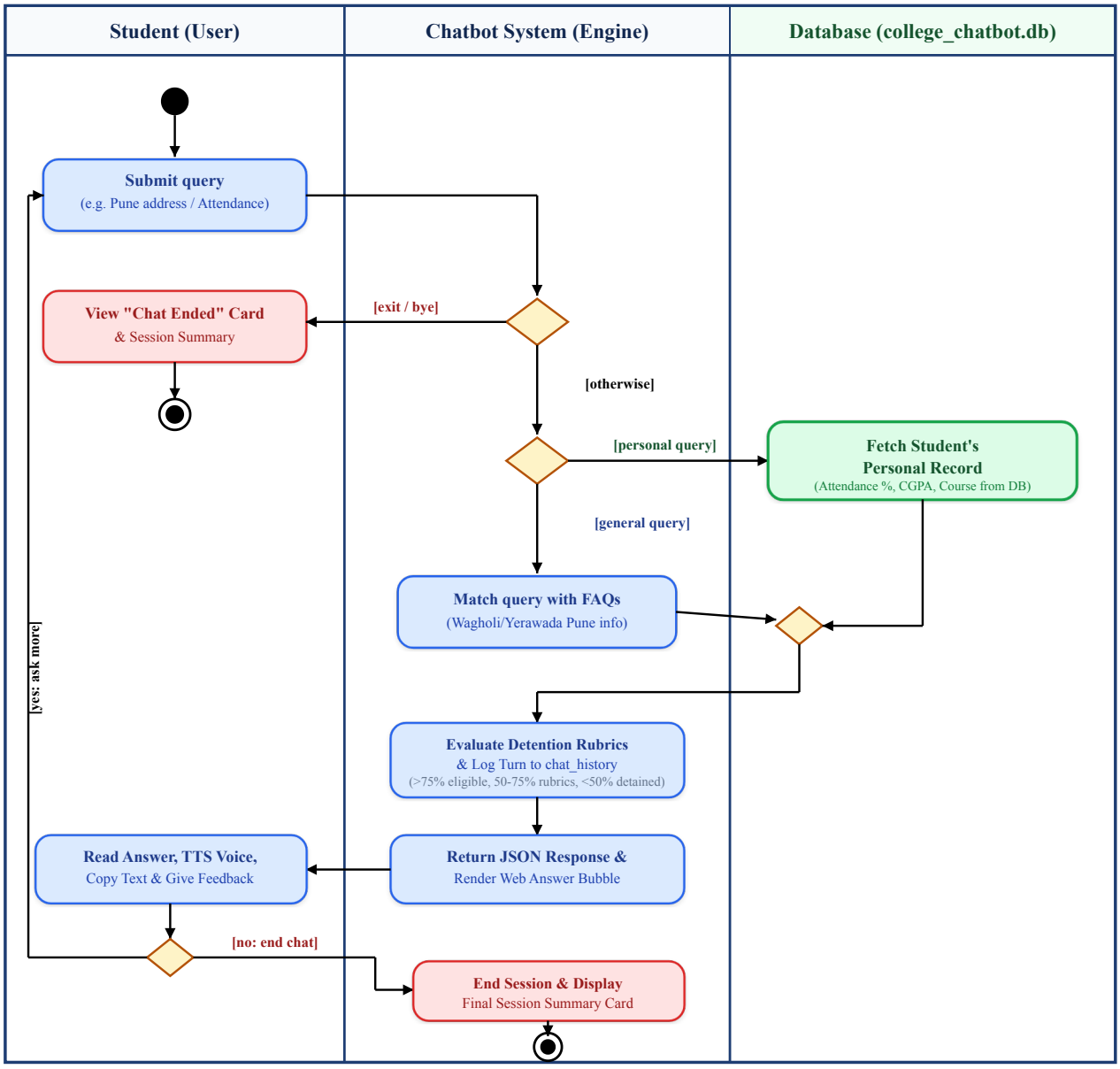
Object Diagrams | Unified Modeling Language (UML) ⚙ GfG

SEQUENCE DIAGRAM :

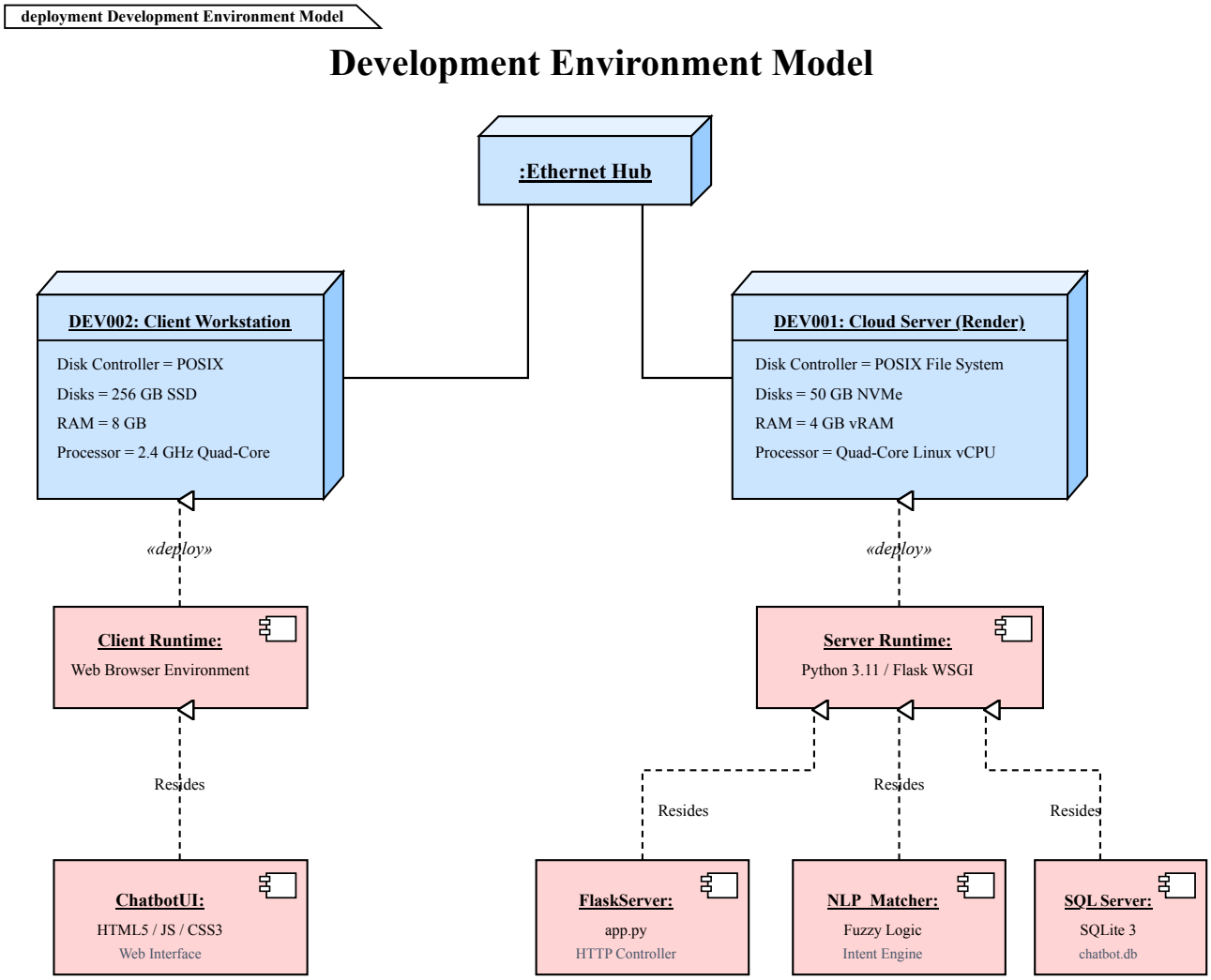


[✕ Session Formally Terminated & Lifelines Closed]

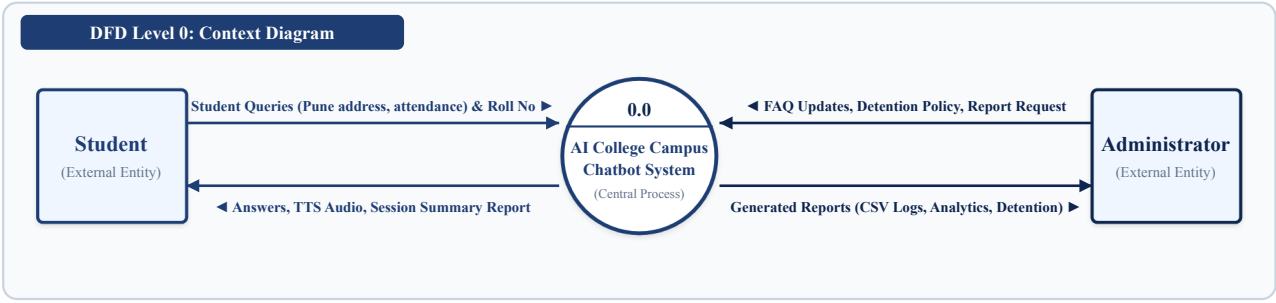
ACTIVITY DIAGRAM :

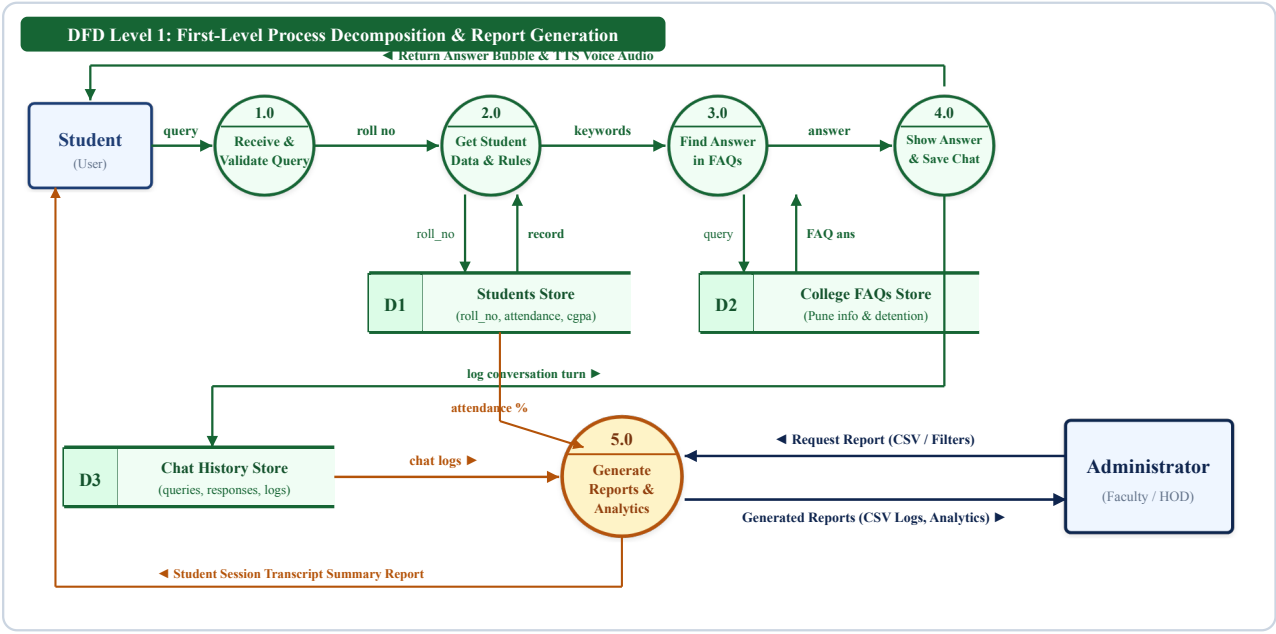


COMPONENT DIAGRAM :



DATA FLOW DIAGRAM :

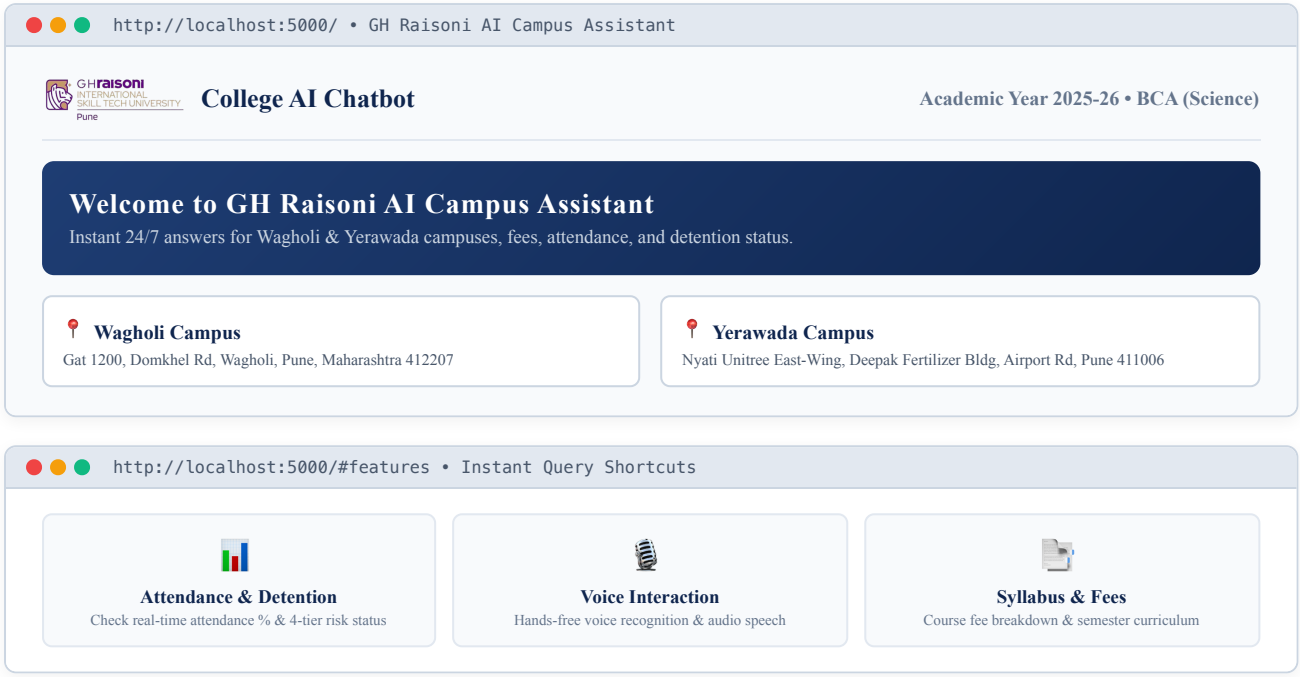




Standard DFD Notation & Symbol Conventions (Gane-Sarson & DeMarco)

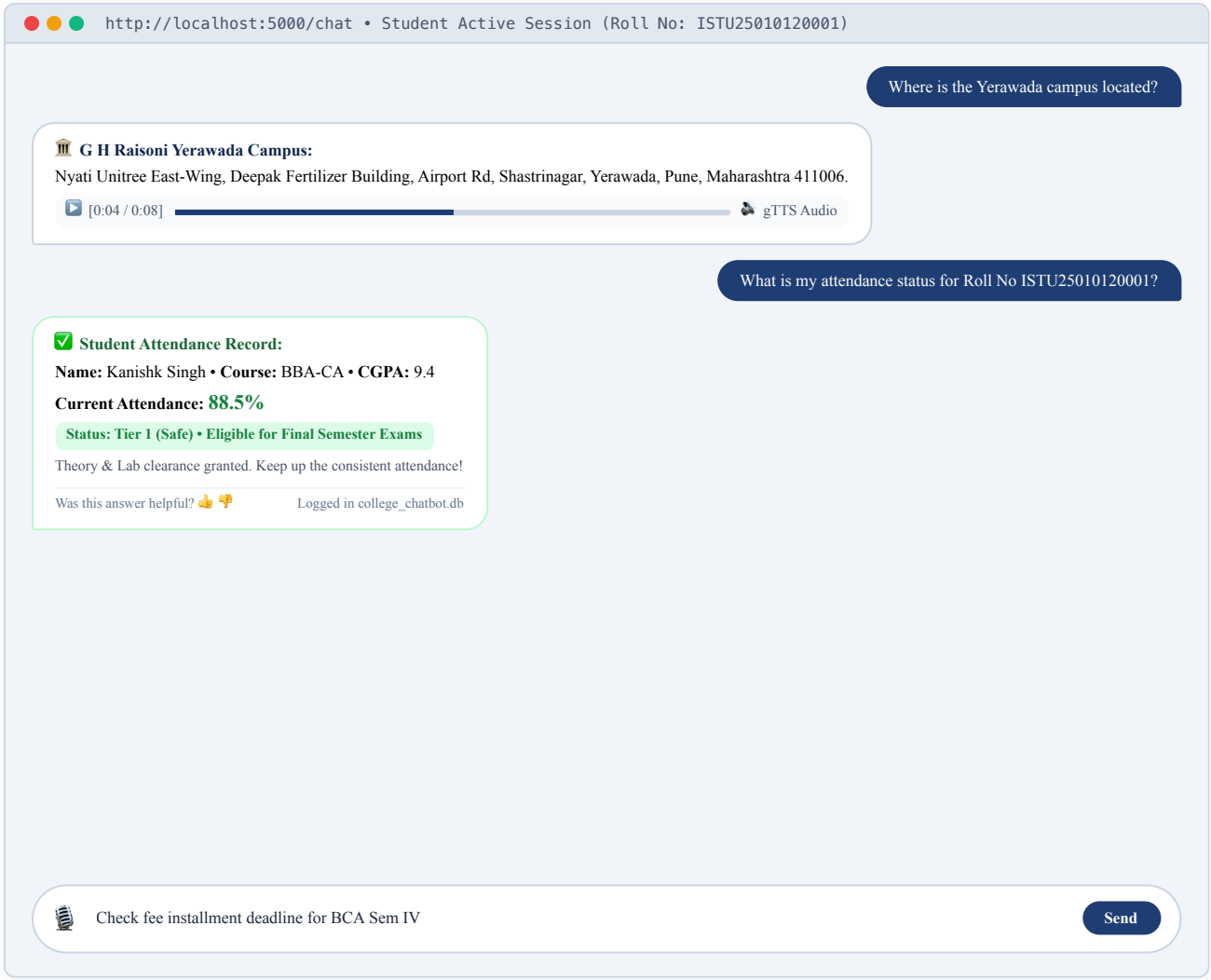
Symbol Shape	DFD Element	Standard Academic Definition	Implementation in Our Project
▭	**External Entity**	Origin/destination of data outside boundary (Square)	**Student (User) & Administrator (Faculty/HOD)**
◯	**Process (Bubble)**	Transforms input data into output data (Circle with ID) Calculates, retrieves, formats, or generates reports	**1.0 to 4.0 Query Handlers & 5.0 Generate Reports** Query answering & administrative CSV/analytics reporting
D ▭	**Data Store** *(Open-Ended)*	Repository of data at rest (Open parallel lines) **MANDATORY RULE: Right side must be open; NEVER closed**	**D1 Students, D2 College FAQs, D3 Chat History** Persistent SQLite tables feeding search & report generation
→	**Data Flow**	Directed solid arrow showing data path & payload Transfers packets between entities, processes, and stores	**query, record, FAQ answer, generated reports CSV** Directional vectors with explicit noun-phrase labels

HOME SCREEN :


http://localhost:5000/ • GH Raisoni AI Campus Assistant
College AI Chatbot
Academic Year 2025-26 • BCA (Science)
Welcome to GH Raisoni AI Campus Assistant
Instant 24/7 answers for Wagholi & Yerawada campuses, fees, attendance, and detention status.
Wagholi Campus
Gat 1200, Domkhel Rd, Wagholi, Pune, Maharashtra 412207
Yerawada Campus
Nyati Unitree East-Wing, Deepak Fertilizer Bldg, Airport Rd, Pune 411006

http://localhost:5000/#features • Instant Query Shortcuts
Attendance & Detention
Check real-time attendance % & 4-tier risk status
Voice Interaction
Hands-free voice recognition & audio speech
Syllabus & Fees
Course fee breakdown & semester curriculum

STUDENT PORTAL & VOICE CHAT MODULE :



ADMIN COMMAND CENTER & REPORT GENERATION :

http://localhost:5000/admin • Central Administrative Telemetry Dashboard

1,482	450	18	99.98%
Total Inquiries	Active Students	Detention Alerts	System Uptime

Administrative Analytical Exports: Export Chat Logs (.CSV) | Export Detention Summary (.CSV) | Add New FAQ

Recent Student Activity & Feedback Audit

Roll No	Query	Category	Attendance	Detention Tier	Feedback
ISTU25010120001	Where is Yerawada campus?	Campus Location	88.5%	Safe (Tier 1)	Helpful
ISTU25010120002	Check my attendance %	Attendance	82.0%	Safe (Tier 1)	Helpful
ISTU0005	Detention status check	Detention	54.5%	Critical (Tier 3)	Helpful
ISTU0006	Exam eligibility query	Detention	42.0%	Severe (Tier 4)	Helpful

TESTING AND IMPLEMENTATION :

TESTING STRATEGIES:

1. Top-Down Strategy: Testing begins from top-level user interface forms and navigation routes, verifying user inputs and simulated mock responses before integrating database models.

2. Bottom-Up Strategy: Low-level database procedures, SQLite schema constraints, and gTTS audio helper functions are unit-tested independently prior to integration into Flask endpoints.

3. Incremental Testing Strategy: Features are implemented and validated step-by-step (e.g., student login → keyword search → attendance lookup → detention logic → voice audio synthesis).

4. System Testing Strategy: The entire integrated web application is tested holistically under real-world scenarios to verify response latency, concurrent student sessions, and CSV report export integrity.

TESTING TECHNIQUES:

1. Black Box Testing: Functionality is validated strictly against functional requirements without inspecting internal code logic. Techniques include Equivalence Partitioning and Boundary Value Analysis on attendance cutoffs (75.0%, 74.9%, 60.0%, 59.9%, 50.0%, 49.9%).

2. White Box Testing: Internal code structure, execution paths, SQL parameterized statements, and regex keyword matcher loops are verified for branch coverage and error handling.

Test ID	Test Description	Test Input	Expected Output	Status
TC_01	Student Login	Roll No: ISTU25010120001	Profile Loaded (Kanishk Singh)	**PASS**
TC_02	Yerawada Address	"Where is Yerawada?"	Nyati Unitree East-Wing Address	**PASS**
TC_03	Attendance Check	"Check attendance"	88.5% • Tier 1 (Safe)	**PASS**
TC_04	Critical Detention	Roll No with 54.5%	Tier 3 Warning • Parent Hearing	**PASS**
TC_05	Voice Synthesis	Click Audio Button	gTTS .mp3 stream generated	**PASS**
TC_06	CSV Export	Admin Click Export	RFC 4180 CSV File Downloaded	**PASS**

DRAWBACKS AND LIMITATIONS :

While the College AI Chatbot provides comprehensive, real-time campus assistance and automated detention governance, like any software system, it operates under certain inherent technical and operational limitations:

1. Dependency on Predefined Knowledge Base :
The chatbot utilizes regex keyword matching and relational FAQ indexing. If a student query uses highly non-standard slang, complex colloquial phrasing, or concerns a novel topic not registered in the SQLite knowledge base, the system falls back to a default guidance response rather than synthesizing new answers through generative AI.

2. Client-Side Speech Recognition Vulnerability :
The voice input feature relies upon the browser Web Speech API. Ambient background noise in crowded campus corridors or poor microphone quality can lead to transcription errors, requiring the student to re-speak or revert to keyboard typing.

3. Single-Tenant SQLite Database Concurrency :
The system currently deploys SQLite 3 as its primary persistence engine. While exceptionally fast for read-heavy operations, SQLite employs file-level database locking during write operations, which may create a performance bottleneck if thousands of students concurrently submit chat evaluations.

4. Internet Connectivity Requirement for gTTS :
While FAQ matching executes locally on the Python Flask server, synthesizing new audio files via Google Text-to-Speech (gTTS) requires outbound internet access. In the event of network disruption, voice output falls back to browser-native synthesis or text-only responses.

PROPOSED ENHANCEMENTS :

To build upon the successful implementation of the College AI Chatbot, several progressive architectural enhancements are planned for future versions:

1. Multilingual Support (Marathi, Hindi & English) :
Recognizing the regional demographic diversity across Pune and Maharashtra, the natural language understanding engine can be expanded to support Marathi and Hindi conversational queries and spoken voice responses, greatly enhancing accessibility for parents and rural students.

2. Local Open-Source LLM Integration (LLaMA / Gemini API) :
Upgrading the query matcher with a Retrieval-Augmented Generation (RAG) architecture powered by an on-premise Large Language Model (e.g. LLaMA 3.2 or Gemini Flash API) will allow the chatbot to answer nuanced, multi-part academic questions by directly referencing institutional PDF handbooks.

3. Automated Guardian SMS & WhatsApp Alerts :
Integrating with institutional SMS/WhatsApp gateways (e.g., Twilio or Gupshup) to automatically dispatch early-warning attendance alerts directly to parents whenever a student enters Tier 2 (Warning) or Tier 3 (Critical) status.

4. Enterprise Database Migration & Native Mobile App :
Migrating the backend persistence layer from SQLite to PostgreSQL for high-concurrency scaling, and wrapping the frontend in a cross-platform Flutter/React Native application for Android and iOS devices.

CONCLUSION :

The **College AI Chatbot** (GH Raisoni AI Campus Assistant) has been conceptualized, engineered, and validated as an effective conversational AI solution addressing critical information distribution challenges at **G H Raisoni International Skill Tech University, Pune**.

The platform successfully bridges communication barriers between students and university departments. By automating routine inquiries regarding **Wagholi and Yerawada campus locations**, semester fee schedules, academic syllabi, and administrative procedures, the chatbot significantly mitigates administrative desk congestion, allowing staff to dedicate their attention to higher-order academic governance.

A standout accomplishment is the implementation of the **4-Tier Attendance Detention Engine**. By providing verified students with real-time percentage evaluations and institutional remedial directives early in the semester, the system replaces anxiety and end-of-term surprises with actionable academic guidance. Students in good standing (≥75%) receive immediate exam reassurance, while those at risk are alerted in time to complete makeup assignments or attend counseling.

The inclusion of multimodal voice recognition and Google Text-to-Speech playback delivers an inclusive, accessible user experience. Coupled with an administrative command center providing telemetry and one-click CSV report exports, the system establishes a robust technological foundation for next-generation digital campus operations at G H Raisoni International Skill Tech University.

BIBLIOGRAPHY

1. Sommerville, I. (2015). *Software Engineering* (10th Edition). Pearson Education.
2. Pressman, R. S., & Maxim, B. R. (2019). *Software Engineering: A Practitioner's Approach* (9th Edition). McGraw-Hill Education.
3. Beck, K. (2002). *Test-Driven Development: By Example*. Addison-Wesley Professional.
4. Booch, G., Rumbaugh, J., & Jacobson, I. (2005). *The Unified Modeling Language User Guide* (2nd Edition). Addison-Wesley.
5. Grinberg, M. (2018). *Flask Web Development: Developing Web Applications with Python* (2nd Edition). O'Reilly Media.
6. Yourdon, E. (1989). *Modern Structured Analysis*. Prentice Hall.
7. Myers, G. J., Sandler, C., & Badgett, T. (2011). *The Art of Software Testing* (3rd Edition). John Wiley & Sons.
8. IEEE Computer Society. (2024). *SWEBOK: Guide to the Software Engineering Body of Knowledge* (Version 4.0). IEEE.
9. W3C Speech API Community Group. (2023). *Web Speech API Specification*. World Wide Web Consortium.
10. McConnell, S. (2004). *Code Complete: A Practical Handbook of Software Construction* (2nd Edition). Microsoft Press.